Line Parameters III

Turn charge and conductor geometry into shunt susceptance and charging Mvar

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Lecture outline

A line can draw current even with its receiving load disconnected. The series $R + \mathrm{j}X$ model does not explain that current.

1. Charge and field: connect conductor voltage to stored charge.
2. Phase capacitance: convert field geometry into charge per volt.
3. Charging: turn capacitance into current and reactive power.
4. Exercises: build and interpret the complete shunt record.

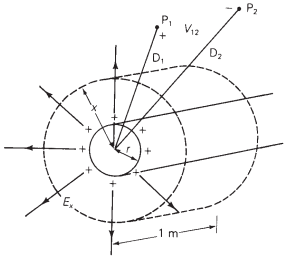
Explain and calculate charging current from line geometry, voltage, frequency, and length.

PART 1

Charge, field, and voltage

An energized conductor stores surface charge. Find how its electric field creates a voltage difference before defining capacitance.

Charged conductor



Positive surface charge

Field points outward from positive charge; negative charge reverses it. The electric field occupies the space between conductor surfaces.

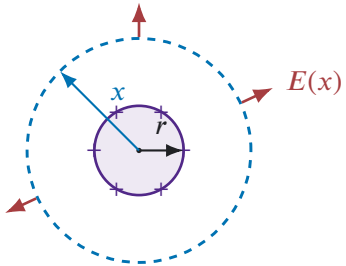
Capacitance describes how much charge is needed to establish a voltage difference.

The electric field begins at charge. Magnetic field lines encircle current.

Source: Glover et al., sixth edition (2017), Fig. 4.18 and Sec. 4.8.

Electric-field quantities

Use a one-metre slice of a long, uniformly charged conductor.



Gaussian cylinder: cross-section

curved area $= 2\pi x\,(1\text{ m})$

q': charge per length (C/m). A one-metre slice encloses $q'(1\text{ m})$.

r: physical radius (m); $x > r$: observation radius (m).

ϵ: permittivity (F/m); in air, $\epsilon \approx \epsilon_0 = 8.85419 \times 10^{-12}$ F/m.

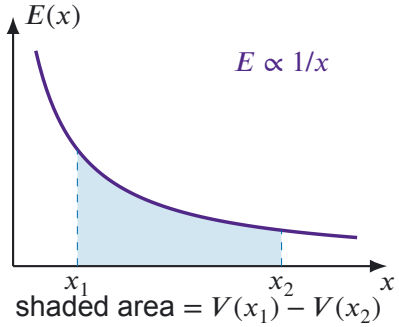
Gauss's law, with zero flux through the end caps:

$$E(x)\,2\pi x(1\text{ m}) = \frac{q'(1\text{ m})}{\epsilon}.$$

The same enclosed charge spreads its electric flux over a larger cylindrical surface as x increases.

Field and potential difference

For positive q', the outward field decreases with radial distance $x > r$.



$$E(x) = \frac{q'}{2\pi\epsilon x} \quad \text{[V/m]}.$$

Voltage difference is the area under $E(x)$:

$$V(x_1) - V(x_2) = \int_{x_1}^{x_2} E(x)\,dx$$

$$= \frac{q'}{2\pi\epsilon} \ln \frac{x_2}{x_1}.$$

Predict: for the same small outward step, is the voltage change larger near the conductor or farther away?

Source: Glover et al., sixth edition (2017), Sec. 4.8.

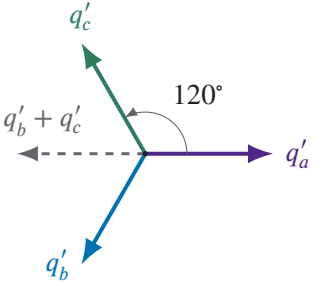
PART 2

Phase capacitance

The field calculation gives potential differences between points. A balanced network needs one phase-to-neutral charge–voltage relation.

Balanced phase charge

q'_a, q'_b, q'_c are charge phasors per unit length; equal magnitudes are separated by $120°$.



phasor plane, not physical positions

The two sloping phasors sum to $-q'_a$:

$$q'_a + q'_b + q'_c = 0.$$

Choose the neutral potential reference n so that

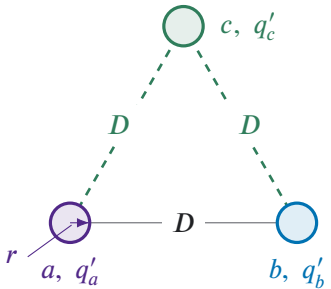
$$V_{an} + V_{bn} + V_{cn} = 0.$$

These are phase-voltage RMS phasors. Equal spacing or complete transposition gives a common phase charge-to-voltage ratio.

A zero sum does not mean each charge is zero. The neutral potential reference does not require a physical neutral wire.

From individual charges to line voltage

Equal centre spacing $D \gg r$, identical radii, negligible earth effect. Compare $V_a - V_b$.



Each source contributes a distance ratio:

$$V_{ab}^{(a)} = \frac{q'_a}{2\pi\epsilon} \ln \frac{D}{r},$$

$$V_{ab}^{(b)} = \frac{q'_b}{2\pi\epsilon} \ln \frac{r}{D},$$

$$V_{ab}^{(c)} = \frac{q'_c}{2\pi\epsilon} \ln \frac{D}{D} = 0.$$

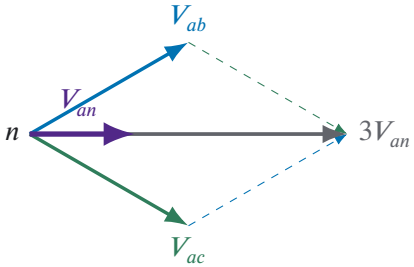
$$V_{ab} = \frac{q'_a - q'_b}{2\pi\epsilon} \ln \frac{D}{r}$$

Explain from the geometry: why does c contribute zero to V_{ab} even though its electric field is present?

From line voltage to phase capacitance

By symmetry, $V_{ac} = (q'_a - q'_c)\ln(D/r)/(2\pi\epsilon)$.

phasor addition



The balanced phase voltages and charges each sum to zero:

$$\begin{aligned} V_{ab} + V_{ac} &= 2V_{an} - V_{bn} - V_{cn} \\ &= 3V_{an}, \end{aligned}$$

$$2q'_a - q'_b - q'_c = 3q'_a,$$

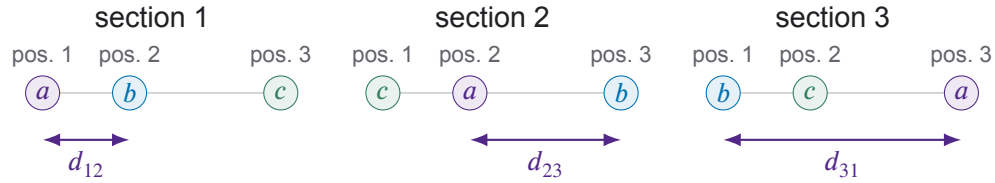
$$V_{an} = \frac{q'_a}{2\pi\epsilon} \ln \frac{D}{r}.$$

$$C'_n = \frac{q'_a}{V_{an}} = \frac{2\pi\epsilon}{\ln(D/r)}$$

The factors of three cancel. C'_n is the phase-to-neutral capacitance rate (F/m).

Transposed phase spacing

Cross-sections of three equal route sections, each of length $\ell/3$. Track a and b across the fixed, unequal position spacings.



$$\frac{\ln(d_{12}/r) + \ln(d_{23}/r) + \ln(d_{31}/r)}{3} = \ln \frac{(d_{12}d_{23}d_{31})^{1/3}}{r}.$$

$$\boxed{D_m = (d_{12}d_{23}d_{31})^{1/3}} \qquad \text{geometric mean distance (GMD)}$$

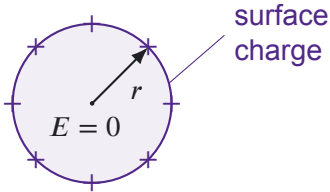
Every phase pair samples all three spacings. Averaging voltage-per-charge coefficients gives one common phase model.

Source: Glover et al., sixth edition (2017), Sec. 4.10.

Phase capacitance rate

One conductor per phase; balanced, completely transposed geometry; negligible earth effect.

$$C_n' = \frac{2\pi\epsilon}{\ln(D_m/r)} \quad \text{[F/m]}$$



ideal electrostatic conductor

The electric self distance begins at the physical surface, so use outside radius r.

Magnetic GMR, D_s, incorporates internal magnetic linkage. That correction belongs in the inductance model.

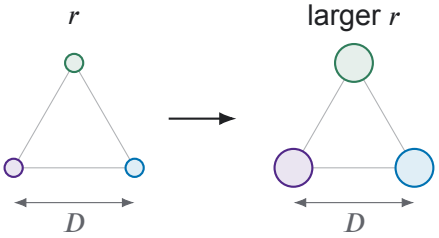
C_n' is a rate. Route length has not yet entered.

Conductor geometry and dielectric permittivity determine charge stored per volt, per unit length.

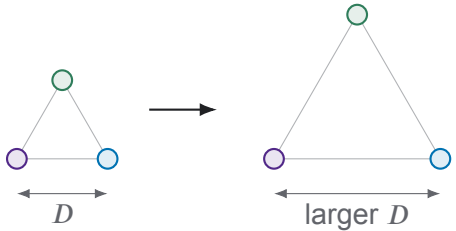
Geometry and capacitance

$$C_n' = \frac{2\pi\epsilon}{\ln(D/r)} \qquad \text{equal spacing: } D = D_m$$

Increase radius; keep spacing fixed



Increase spacing; keep radius fixed



Predict each change in C_n'. At the same phase voltage, which geometry stores more charge? Use $\ln(D/r)$ to check.

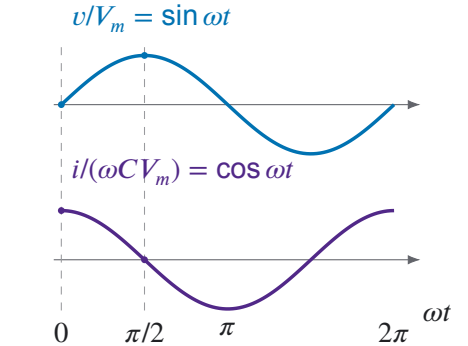
PART 3

Charging current and power

Capacitance tells us how much charge is stored. AC voltage keeps changing that charge, so we now need a shunt-current model.

Susceptance per unit length

AC voltage continuously changes the stored charge, $q = Cv$.



normalized waves; V_m: voltage peak

Time relation and RMS phasors:

$$i_C = \frac{dq}{dt} = C\frac{dv}{dt},$$

$$I_C = \mathrm{j}\omega C V, \quad \omega = 2\pi f,$$

$$Y' = \mathrm{j}\omega C'_n = \mathrm{j}B'_{\mathrm{sh}}.$$

$$\boxed{B'_{\mathrm{sh}} = 2\pi f C'_n} \quad [\mathrm{S/m}].$$

Primed quantities are per unit length.
Neglect G'; capacitive $B'_{\mathrm{sh}} > 0$.

Predict i_C at peak voltage and at the rising voltage zero crossing.

Route shunt totals

Capacitance and susceptance totals

$$C_n = C_n' \ell$$

$$B_{\text{sh}} = B_{\text{sh}}' \ell = \omega C_n$$

$$Y = \text{j} B_{\text{sh}}$$

Compatible units

$$1 \text{ pF/m} = 1 \text{ nF/km}$$

$$C_n \text{ [nF]} = C_n' \text{ [nF/km]} \; \ell \text{ [km]}$$

$$B_{\text{sh}} \text{ [}\mu\text{S]} = B_{\text{sh}}' \text{ [}\mu\text{S/}km\text{]} \; \ell \text{ [km]}$$

Multiply a rate by length once. Check the total using both $B_{\text{sh}}' \ell$ and ωC_n.

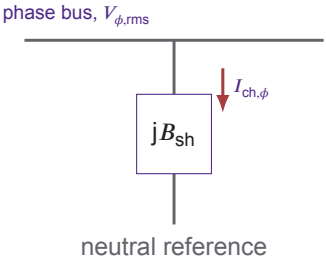
Charging current

A bus rating is normally line-to-line:

$$V_{\phi,\text{rms}} = \frac{V_{LL,\text{rms}}}{\sqrt{3}}.$$

The per-phase shunt current magnitude is

$$|I_{\text{ch},\phi}| = B_{\text{sh}} V_{\phi,\text{rms}}.$$



Using $V_{LL,\text{rms}}$ directly in the per-phase current formula makes the current too large by $\sqrt{3}$.

Capacitive reactive power

With current entering the capacitor, use absorbed complex power $S = VI^*$. V_ϕ is a phase-voltage phasor; V_{LL} is the line-to-line RMS magnitude:

$$I_C = \mathrm{j}B_{\mathrm{sh}}V_\phi, \qquad B_{\mathrm{sh}} = \omega C_n > 0.$$

For three balanced phases,

$$S_C = 3V_\phi I_C^* = -\mathrm{j}3B_{\mathrm{sh}}|V_\phi|^2 = -\mathrm{j}B_{\mathrm{sh}}V_{LL}^2.$$

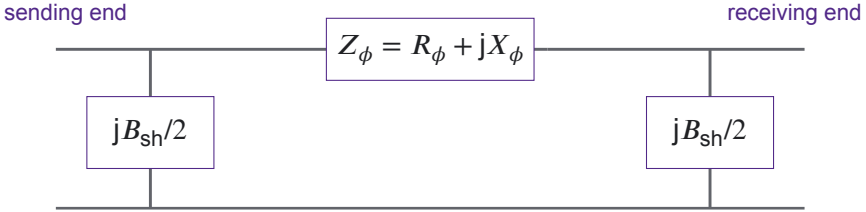
Thus the signed absorbed reactive power and supplied magnitude are

$$Q_C = B_{\mathrm{sh}}V_{LL}^2 > 0, \qquad S_C = -\mathrm{j}Q_C.$$

Following the textbook, Q_C denotes positive var supplied by capacitance. Its absorbed reactive power is $-Q_C$.

Nominal-π shunt split

Place the calculated shunt quantity in a usable terminal circuit

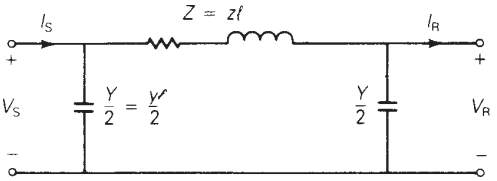


total route shunt: $\mathrm{j}B_{\mathrm{sh}}/2 + \mathrm{j}B_{\mathrm{sh}}/2 = \mathrm{j}B_{\mathrm{sh}}$

The nominal-π approximates distributed charging with two end shunts. Each uses its local phase voltage to calculate current; together they retain the total route admittance.

An unloaded energized line

V_R is receiving phase-to-neutral voltage; I_R is receiving load current; $I_{\text{sh},R}$ enters the receiving shunt. Y is the total per-phase route shunt admittance.



Load disconnected

$$I_R = 0, \qquad I_{\text{sh},R} = \frac{Y}{2} V_R$$

An open receiving load leaves the line capacitance connected to the energized conductors.

The source still supplies charging current.

Zero load current does not imply zero current everywhere along the line.

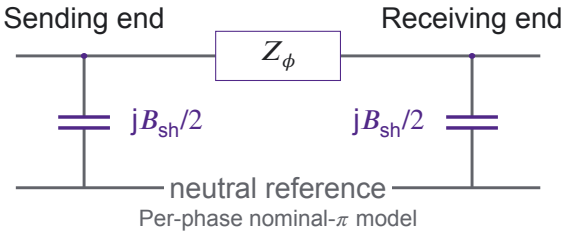
Source: Glover et al., sixth edition (2017), Fig. 5.3; Secs. 4.10 and 5.1.

PART 4

Exercises

We have the path from geometry to capacitance and charging. Use it to calculate a route record and explain current at no external load.

Exercise 1: given and find



Per-phase nominal-π model

Balanced-line data

$$B_{sh} = 0.400 \text{ mS},$$
$$V_{LL} = 138 \text{ kV}.$$

B_{sh} is full-route, per phase.

Use the same line-voltage magnitude at both ends.

1. Find the total three-phase capacitive Mvar supplied.
2. Find phase charging current and each nominal-π endpoint Mvar.
3. Predict the supplied Mvar at 0.95 of this voltage.

Exercise 1: solution

$$Q_C = B_{\text{sh}} V_{LL}^2 = 0.000400(138)^2 = 7.6176 \text{ Mvar}$$

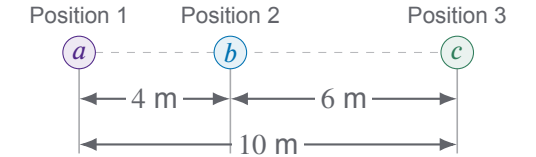
$$|I_C| = B_{\text{sh}} \frac{V_{LL}}{\sqrt{3}} = 31.87 \text{ A}$$

$$Q_{C,\text{end}} = 3.8088 \text{ Mvar}, \qquad Q_C(0.95V) = 0.95^2(7.6176) = 6.8749 \text{ Mvar}$$

Q_C is supplied magnitude. Signed absorption is $-Q_C$. The endpoint halves sum to the total only at equal terminal voltages.

Exercise 2: given

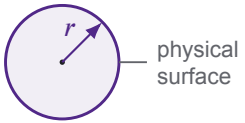
Fully transposed line; one conductor per phase



One transposition section; center-to-center distances

$$V_{LL} = 138 \text{ kV}, \qquad f = 60 \text{ Hz},$$
$$\ell = 75.0 \text{ mi} = 120.7008 \text{ km},$$
$$\epsilon = \epsilon_0 \approx 8.8541878 \times 10^{-12} \text{ F/m}.$$

Conductor cross-section



Enlarged; not to scale

$$r = 14.00 \text{ mm} = 0.01400 \text{ m}$$

GMR (comparison only):

$$D_s = 10.90 \text{ mm}.$$

Model: Three equal transposition sections; air; negligible earth effect.

Exercise 2: find

Find and verify

(a) Find C_n' in nF/km using phase spacings 4, 6, and 10 m and the physical radius.

(b) Find B_{sh}' in μS/km, then route C_n and B_{sh}.

(c) At uniform 138-kV line-to-line voltage, calculate phase charging current, total supplied Mvar, and each shunt half.

(d) Predict the effects of doubling length and of substituting the wrong radius or mean spacing.

State the units and model assumptions with each result.

Ex. 2(a): electric ratio

$$\frac{D_m}{r} = \frac{6.21447\ \text{m}}{0.01400\ \text{m}} \approx 443.890358,$$

$$\boxed{\ln\left(\frac{D_m}{r}\right) \approx 6.095577590.}$$

The ratio is dimensionless and greater than one. The numerator is transposed phase-center spacing. The denominator is physical surface radius.

Using 10.90 **mm GMR would answer a magnetic-self question inside an electric-field formula.**

Ex. 2(a): capacitance rate

$$C_n' = \frac{2\pi(8.85419 \times 10^{-12})}{6.09558} \text{ F/m},$$

$$\boxed{C_n' \approx 9.126699143 \text{ pF/m}}$$

$$\boxed{C_n' \approx 9.126699143 \text{ nF/km}.}$$

The numerical value is unchanged from pF/m to nF/km because both numerator and denominator units scale by one thousand.

Ex. 2(b): susceptance rate

$$B'_{\text{sh}} = 2\pi f C'_n$$

$$B'_{\text{sh}} = 2\pi(60)(9.1267 \times 10^{-9}) \text{ S/km},$$

$$\boxed{B'_{\text{sh}} \approx 3.440684518 \times 10^{-6} \text{ S/km}}$$

$$\boxed{B'_{\text{sh}} \approx 3.440684518 \ \mu\text{S/km}.}$$

At fixed geometry, capacitance is independent of frequency in this model, while susceptance is proportional to frequency.

Ex. 2(b): route totals

75.0 **miles** = 120.7008 **kilometres**

Total phase-to-neutral capacitance

$$\begin{aligned} C_n &= (9.1267\ \text{nF/km}) \\ &\quad \times (120.7008\ \text{km}) \\ &\approx 1101.599888\ \text{nF} \\ &\approx \boxed{1.1016\ \mu\text{F}.} \end{aligned}$$

Total shunt susceptance

$$\begin{aligned} B_{\text{sh}} &= (3.44068\ \mu\text{S/km}) \\ &\quad \times (120.7008\ \text{km}) \\ &\approx \boxed{0.415293\ \text{mS}.} \end{aligned}$$

The route from total C_n through $2\pi f C_n$ returns the same total B_{sh}.

Ex. 2(c): charging quantities

Use line-to-neutral voltage exactly once

Voltage and charging current

$$V_{\phi,\text{rms}} = \frac{138\ \text{kV}}{\sqrt{3}} \approx 79.674337\ \text{kV},$$

$$\boxed{|I_{\text{ch},\phi}| = B_{\text{sh}} V_{\phi,\text{rms}} \approx 33.088224\ \text{A}.}$$

Mvar and nominal-π split

$$\boxed{|Q_{\text{ch},3\phi}| = B_{\text{sh}} V_{LL,\text{rms}}^2 \approx 7.908847\ \text{Mvar}.}$$

$$\boxed{B_{\text{sh}}/2 \approx 0.207646687\ \text{mS}.}$$

per phase at each end.

Use $V_{LL,\text{rms}}/\sqrt{3}$ for phase current. Total Mvar uses total B_{sh}; each end receives $B_{\text{sh}}/2$.

Ex. 2(d): comparisons

Error signatures

- GMR 10.90 mm used as radius: $C' = 8.76672$ pF/m, 3.9442% low.
- arithmetic mean spacing 6.6667 m: $C' = 9.02273$ pF/m, 1.1392% low.
- diameter 28 mm used as radius: $C' = 10.29768$ pF/m, 12.8303% high.

Double-length check: 150 mi

$$C_n = 2.2032\ \mu\text{F},$$
$$B_{\text{sh}} = 0.830587\ \text{mS},$$
$$|I_{\text{ch},\phi}| = 66.1764\ \text{A},$$
$$|Q_{\text{ch},3\phi}| = 15.8177\ \text{Mvar}.$$

At fixed geometry, frequency, and voltage, C, B, charging current, and charging Mvar all scale linearly with route length.

Summary

- **Electric-field geometry:** capacitance uses physical conductor radius and transposed phase GMD. Magnetic GMR belongs to inductance.
- **Network parameter:** capacitance becomes shunt susceptance through $B_{\text{sh}} = \omega C_n$ and scales linearly with route length.
- **Charging quantities:** phase charging current uses $V_{LL,\text{rms}}/\sqrt{3}$. Total three-phase charging var uses $B_{\text{sh}} V_{LL,\text{rms}}^2$.
- **Nominal-π placement:** the total per-phase shunt susceptance is divided equally between the two terminals.

An energized AC line keeps charging even without external load. Geometry determines capacitance; frequency and voltage determine the resulting shunt current.